

An Artificial Intelligence Framework for Dynamic Selection and Resource Allocation for EVs in Vehicular Networks

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Abstract—Wireless power transfer for charging electric vehicles (EVs) using the inductive charging mechanism enables EVs to recharge their batteries wirelessly while in motion or during halts at traffic signals. Additionally, resonant inductive charging (RIC) can wirelessly transfer energy with high efficiency and over longer distances. To achieve these objectives, we propose a RIC-enabled system for traffic signal scenarios capable of selecting EVs based on their required energy needs for travel and the remaining traffic signal time. We formulate an optimization problem that allows the selected EVs to maximize their battery energy during traffic signal halts, consequently optimizing the defined utility function. Given the dynamic nature of the problem, it falls under the category of NP-hard problems, and to address this, we propose a novel artificial intelligence framework. Our approach utilizes both the halt time and energy demand information, resulting in the development of the Traffic Signal-Aware Electric Vehicle Selection and Resource Allocation algorithm. We employ long short-term memory based deep learning model to predict battery energy needs and generate energy demand score information. The generated demand score, along with the remaining traffic signal time, serves as conditions for the final selection and allocation of charging resources to EVs. Finally, experimental results confirm the effectiveness of our proposed method, demonstrating superior performance compared to the deep neural network model. Furthermore, in terms of battery energy, our approach achieves a 55.1% increase compared to baseline-B and an 8.4% increase compared to baseline-C.

Index Terms—Electric Vehicles, Resonant Inductive Charging (RIC), Dynamic Selection and Resource Allocation (DSRA), Vehicular Networks.

I. INTRODUCTION

Today's era is an age of Intelligent systems and machines, where each and every sector of human life is greatly affected by these autonomous systems, and the transportation sector is also a major part of them. Electrical Vehicles (EVs) is a type of automobile that uses electric motors for propulsion instead of fossil fuels while ensure sustainability, reduce the risk of several climate changes. Hence, different countries including United States and European Union have taken numerous steps

to reduce carbon-dioxide emissions by incorporating EVs while reducing the amount of fossil fuel based vehicles [1], [2]. In recent years, the global market of EVs increased a lot and it is predicted that 1.2 million EVs would be sold annually in the United States by 2025 while the amount would be 3 million in China and 0.19 million in South Korea, and 41 million globally by 2040 [3]–[6].

For facilitating the charging process for the increasingly growing number of EVs, a huge number of charging stations (CS) will be required, while a lot of CS has been constructed as well [7], and the high load for charging EVs could also affect the stability of power grid operations [5]. There are mainly two approaches to charge the battery of EVs whereas connecting the vehicle by wire plug with charging poles in the charging stations at work and at home is the traditional one, and the other way of charging is dynamic wireless power transfer (DWPT) through inductive charging [8], [9]. In addition, resonant inductive charging (RIC) for DWPT, is capable of transferring energy at a high rate, which is greater than 100 kWh, and to a greater distance efficiently; even the sending and receiving coils are not aligned properly [10].

Due to the limited capacity of EV batteries, it is needed to recharge frequently according to the use of EVs and for some cases, like EV taxis, which need to travel a lot, it is multiple times in a day [11]. So, managing the energy demand and energy supplement according to the EVs need should be investigated with proper concern for effective performance output from the EVs [11]–[13]. In the last decade, dynamic charging technology got much attention for the charging of EVs and it is possible to recharge the battery of EVs while vehicles are in running state or stopping in a traffic signal [8]–[15]. In work [16], cost minimization for the deployment of wireless charging systems in possible candidate road locations inside metropolitan areas has been analyzed depending on the vehicle traffic. An optimization problem is formulated to determine the optimal charging lanes for WPT while considering the mobility of vehicles and classifying the vehicles according to road traffic. The authors of [17] proposed a system for determining the charging area considering the multi-excitation unit based on the required output power by calculating mutual inductance for dc current input at different times. The system is capable of improving its power capacity without raising voltage or current stress.

In [18], a nearly optimal dynamic wireless charging strategy

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depending on adaptive learning strategy has been proposed using dynamic programming. Here, a feedback-based optimization technique with continuous state space and action space is considered, where the feedback action ensures uniformity with respect to the initial operating conditions. For path selection and charging/discharging optimization of EVs based on V2G, the work in [19], designed two optimization methods. They employed the MILP method to identify the best charging plan for the BEV to satisfy the needs of the transportation system's passengers and mixed-integer linear programming (PHEV) to decide the charging schedule to avoid overloading the power grid. An interesting method has been proposed by the authors of [20] to detect the DWPT coil embedded in the road where the coil can be activated instantly for transferring the power. In order to detect EVs driving at high speeds, this research suggests a low-cost and low-power EV detection system based on a new orthogonal coil configuration.

Some existing works [21], [22] designed the optimization process of the dynamic charging system for minimizing the overall cost of EV considering path selection, battery capacity, traffic information's whereas the authors in [23] considered the security and privacy issues of EV drivers and WPT systems. Though several researchers have designed their optimization approach for dynamic wireless charging systems, no existing work has considered the selection of EVs for charging, and in a current situation, all the EVs will not require battery energy. In addition, the energy-need importance of EVs with lower battery status is greater than the energy-need importance of EVs with high battery status. Moreover, the energy supply capacity of the power grid at the present time is also limited, and excessive load could increase the overload risk of the grid [4], [5]. Hence, for proper distribution of energy to the EVs according to need and for effective performance output of the overall process of dynamic wireless charging, it is required to efficiently select EVs and allocate the charging resources.

In our work, we considered traffic signal situations to design our proposed system model, and it is almost impossible for an EV to avoid the signal situations for travelling, especially in urban areas. Vehicles lose a major portion of their travel time during a journey from one place to another, and hence, we proposed a novel system to utilize this wasted time for charging the EVs by intelligently selecting the EVs depending on energy demand and signal time by means of RSU as base station. The brief descriptions of the overall process will be explained in the system model part placed in the next section of this paper. The vital contributions of this work is summarized as follows:

- At first, we investigate the influence of traffic situations and the dissipation of time in the traffic signals during journey of EVs in vehicular network while studying possible advantages of utilizing this loss time to tackle challenges associated with time and energy constraints.
- We formulated an optimization problem for maximizing the amount of achievable battery energy during traffic halt maximizing the energy demand score of the EVs so that EVs can be selected for resource allocation.

- To solve the formulated optimization problem, we propose Long-Short Term Memory (LSTM) based deep learning scheme for predicting energy need information and generating energy demand score. Consequently, we use demand score and remaining signal time as selection criteria for selecting suitable set of EVs from all the available EVs.
- Finally, experimental results demonstrate that our proposed AI framework demonstrates better performance compared to the baseline methods and EVs can achieve on average 55.1% and 8.4% more battery energy than baseline-B and baseline-C, respectively.

II. SYSTEM MODEL

In this section, we explain the architecture of our proposed model and described the associated components related to our system model. Also, an optimization problem is formulated to maximize the obtainable battery energy in a traffic halt environment during journey from source to destination in the vehicular network.

A. System Overview

We design an intelligent framework to determine a suitable subset of EVs automatically from a set of EVs in current situation depending on the traffic signal time and energy demand information of EVs, which is depicted in Fig. 1. After selecting the EVs according to the above mentioned criteria, the vehicles will be allocated for automatic charging through charging pads using RIC mechanism, which are embedded with the road. As shown in Fig. 1, the layout of the road portion for WPT system in the traffic signal area is designed in such a way that can instruct and facilitate the vehicle drivers for aligning with the system structure [24]. In our work, the main aim is to find out the EVs which are in need of charging at present traffic signal situation according to battery energy need information, so that depending on the travel information and EVs information resources can be allocated to the selected EVs. Concretely, a dynamic strategy is implemented to pick out proper vehicles and assign the charging resources to those vehicles.

B. Model Description

We consider a diverse set of V number of EVs denoted by $\mathcal{V} = \{1, \dots, v, \dots, V\}$ and by accepting the request from base station for dynamic WPT process, every vehicle will share several information's using 5G/Wi-Fi wireless communication [21], [26]. The information includes unique identification as EV ID, energy status when departed from source, current energy status, source distance, destination distance, maximum speed, maximum battery capacity and arrival time as I_v , e_v^{depart} , $e_{v,r}^{arrive}$, dis_v^{source} , dis_v^{destin} , Sp_v^{max} , E_v^{max} , $t_{v,r}^{arrive}$ upon arrival in the signal node r . And, the communication process to the RSU will incur a small amount of energy cost as $e_{v,r}^{com}$ and time cost as $t_{v,r}^{com}$. In our proposed system, a RSU will act as a base station with a server of high computation and communication power in this scenario and responsible for the main communication and computation task for the

$E_{d,r}^{max}$ of the RSU node and hence, the EVs will be allocated charging resources while satisfying this two factors.

3) *Waiting time in the Traffic node:* We consider that after arriving to a traffic node and by requesting the resources for allocation every EV have to wait for a certain moment to get the allocation of resources. Let, an EV v ($v \in V$) arrive at a traffic node r ($r \in R$) facilitated with dynamic resource allocation at $t_{v,r}^{arrive}$ time and wait for $t_{v,r}^{wait}$ time and get the resource at $t_{v,r}^{alloc}$. So, the waiting time $t_{v,r}^{wait}$ of the vehicle v can be calculated as,

$$t_{v,r}^{wait} = t_{v,r}^{alloc} - t_{v,r}^{arrive}, \quad (9)$$

where the $t_{v,r}^{alloc}$ can be determined by adding the time of arrival $t_{v,r}^{arrive}$ with the computation time of the RSU server and can be defined as follows:

$$t_{v,r}^{alloc} = t_{v,r}^{arrive} + t_r^{comp}. \quad (10)$$

Also, the total energy consumption and total latency in a traffic node enabled with wireless charging resource allocation for communication and computation mechanism can be calculated as follows [28]:

$$E = e_{v,r}^{com} + e_r^{comp}. \quad (11)$$

$$T = t_{v,r}^{com} + t_r^{comp}. \quad (12)$$

4) *Resource allocation time and Obtainable Energy:* Let, vehicle v at node r get the resources for $t_{v,r}^{ch}$ time and hence, the energy obtained by an EV $e_{v,r}$ in the node r can be evaluated as,

$$e_{v,r} = t_{v,r}^{ch} * pow_{r,n}^{ch}, \quad (13)$$

where $pow_{r,n}^{ch}$ represents the charging power of the charging pad n at the node r and $t_{v,r}^{ch}$ can be found from $t_{v,r}^{ch} = t_{v,r}^{leave} - t_{v,r}^{alloc}$. Here, $t_{v,r}^{leave}$ denotes the time in which the resources are disconnected from an EV. And, the net energy $E_{v,r}$ achieved from a traffic node can be calculated as:

$$E_{v,r} = e_{v,r} - e_{v,r}^{com}. \quad (14)$$

III. PROBLEM FORMULATION

In our work, the main objective is to increase the energy demand score so that EVs able to get the allocation of charging resources and can obtain maximum energy for the battery at a present situation from the RSU assisted and WPT enabled traffic node. For this, we set a utility function (UF) which we want to maximize for the TSTEDRA system. We consider energy demand score, obtainable energy and computation loss in the traffic node to generate the UF and the proposed system performance is proportional to energy demand score and obtainable energy, and inversely proportional to computation loss. So, the UF for our proposed system can be represented as follows:

$$\delta = \eta_{v,r}^{demand} + \zeta_1 E_{v,r} - \zeta_2 L, \quad (15)$$

where $\eta_{v,r}^{demand}$ indicates the energy demand score, $E_{v,r}$ denotes the obtained energy, L represents the loss in the computation process and, ζ_1 and ζ_2 denotes the weighting

ratio for $E_{v,r}$ and L . Therefore, the optimization problem to maximize the utility function for the TSTEDRA system can be formulated as:

$$P1: \max_{C_{v,r}, e_{v,r}^{arrive}, t_{sig}^{thr}} \sum_{v=1}^V \sum_{r=1}^R \delta_{v,r} \quad (16)$$

$$\text{s.t. } C_{v,r} = (0, 1), \forall v \in V, \forall r \in R, \quad (16a)$$

$$t_{sig}^{start} \leq t_{v,r}^{arrive} \leq t_{sig}^{thr}, \quad (16b)$$

$$\forall v \in V, \forall r \in R, \quad (16c)$$

$$e_{v,r}^{com} \leq e_{v,r}^{arrive} \leq e_{v,r}^{thr}, \forall v \in V, \forall r \in R, \quad (16d)$$

$$0 \leq t_{v,r}^{wait} \leq t_{sig}^{thr}, \forall v \in V, \forall r \in R, \quad (16e)$$

$$e_{v,r}^{arrive} < (e_{v,r}^{arrive} + E_{v,r}) \leq E_{v,r}^{max}, \quad (16f)$$

$$n(pow_{r,n}^{ch} * t) \leq E_{d,r}^{max}, \quad (16g)$$

$$e^{min} \leq E \leq e^{max}, \forall v \in V, \forall r \in R, \quad (16h)$$

$$e^{max} \leq \gamma_1, e^{min} \leq \gamma_2, \forall r \in R, \quad (16i)$$

$$f^{min} \leq f \leq f^{max} \quad (16j)$$

In the above problem P1, we consider three decision variables such as $C_{v,r}$ which represents the choice of EVs for getting resources, $e_{v,r}^{arrive}$ which is the battery status of the vehicle v at BS node r and t_{sig}^{thr} i.e. the minimum threshold value of the traffic signal time under which any of the EV v will get the resource for our proposed TSTEDRA system. In our formulated problem P1, constraint 16(a) denotes a binary variable used to set the choice of EVs for getting resources, constraint 16(b) sets the range value for the arrival time of the EVs at BS node, constraint 16(c) limits the values for the current status of battery energy of all EVs, constraint 16(d) ensures the maximum time for resource allocation for EVs, constraint 16(e) assures a minimum waiting time limit for EVs which is from 0 to a threshold value. Also, constraint 16(f) presents the limit for battery status of EVs and ensures that after obtaining energy from RSU node it will not exceed the capacity limit of the battery, constraint 16(g) express the maximum limit of energy delivery capacity of WPT enabled traffic node per unit time t , constraint 16(h) and constraint 16(i) set limits for the total energy consumption value for communication and computation in EVs and RSU server and ensures the value range respectively and the last constraint 16(j) assures a feasible computation frequency for RSU server [28]. The process of EV selection and resource allocation is stochastically distributed over time, hence, the utility function for maximizing problem (P1) directs to a dynamic programming problem that is used for the stochastic environment. As a result, problem (P1) is regarded as an untamable NP-hard problem in deterministic time [30], [31]. So, to achieve

the solution for problem (P1), we decomposed it into two sub-problems which are energy need prediction problem and, EV selection and charging resource allocation problem. And, for solving the sub-problems we have used LSTM model, which is a special type of Recurrent Neural Network (RNN) architecture and the description of the solution approach for the two sub-problems has been illustrated in section IV.

IV. PROPOSED SOLUTION APPROACH BASED ON DEEP LEARNING

Considering the complexity issue of the formulated problem (P1) and since P1 is a NP-hard problem, the solution to the problem requires dynamic learning strategy. Hence, we solve the sub-problems using deep learning model LSTM, which is a type of RNN model and used to process sequential data while avoiding the vanishing gradient problem. We represent the overall solution methodology of the proposed AI framework in Fig. 2.

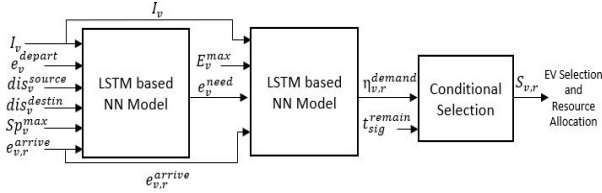


Fig. 2: AI framework of the proposed solution approach for TSTEDRA system.

A. Energy Need Prediction using LSTM

The computation process of our proposed system includes two consecutive parts, whereas the first part is the prediction of the battery energy information for the EVs to reach the destination. We generate this energy need information based on the energy status of EVs during departure e_v^{depart} from source, maximum speed of EVs Sp_v^{max} , energy status at the time of arrival $e_{v,r}^{arrive}$ in BS node, and the distances of source and destination from BS node which are dis_v^{source} and dis_v^{destin} respectively. And for this information prediction task, we utilized a deep learning model based on LSTM which can learn complex patterns from data while considering the dependencies of data from one step to another. Also, LSTM uses memory cells to store and retrieve information over a long period of time, which enables the model to accurately process the inputs for extracting information and to generate effective outputs. In addition, this architecture is very useful to control the updating process of network weights, which is also known as gradients, and can avoid vanishing and exploding gradient problems when gradient values are too small or too large. Moreover, the convergence speed of the LSTM model is fast, which assures a speedy computation for our proposed system that minimizes the waiting time of the EVs at the BS node. We trained and fit our model over the training data while considering the metric MAE as a loss function and adaptive moment estimation (adam) as an optimizer for our model. The loss function MAE calculates the average value of

Algorithm 1 LSTM Model for Battery Energy Need Information Generation for EVs

Input: $e_v^{depart}, e_{v,r}^{arrive}, dis_v^{source}, dis_v^{destin}, e_v^{need}$
 $\forall v \in V, \forall r \in R$

Output: e_v^{need*}

Initialization: LSTM Network Model Parameters

```

1: while ( $Episode \leq Maximum$ ) do
2:   for  $\forall v \in V$  do
3:     if (Training == True) then
4:       Check constraint: Evaluate the constraints (16a),
         (16b), (16c), 16(j)
5:       Evaluate: Mean Absolute Error using (17)
6:       Store: Trained Model Parameters
7:     else
8:       Evaluate:  $e_v^{need*}$  from Trained Model
9:       Store:  $e_v^{need*}$ 
10:    end if
11:  end for
12: end while
    
```

the absolute differences between predicted values and actual values, as shown in the equation below:

$$Loss_{MAE} = \frac{\sum_{k=1}^K |Value_{predict} - Value_{actual}|}{K}. \quad (17)$$

Here, $Loss_{MAE}$ represents the calculated loss, and K presents the number of input samples, where $K = \{1, \dots, k, \dots, K\}$ and $Value_{predict}$ and $Value_{actual}$ denote the predicted and actual values, respectively. The overall procedures for the prediction process of energy needed to reach the destination node for the EVs are depicted in Algorithm 1.

B. EV Selection and Resource Allocation using LSTM and Conditional Selection

In this subsection, we have described the second part of our proposed system workflow, where a subset of EVs will be selected and charging resources will be allocated to the EVs based on the energy need information generated from the previous part and depending on the processed information collected from EVs and traffic infrastructure. To accomplish this task, we have employed a fine-tuned LSTM neural network model like the previous step, and the generated information along with the remaining signal time are used as conditions for selecting and allocating resources to EVs. The deep learning model LSTM is used for generating the energy demand score $\eta_{v,r}^{demand}$ information of the EVs which is shown in equation 7 depending on the maximum battery energy capacity E_v^{max} of the EVs, energy status at the BS node $e_{v,r}^{arrive}$ and battery energy need e_v^{need*} to reach destination. Subsequently, EVs are selected and charging resources are allocated based on both the demand score $\eta_{v,r}^{demand}$ and the remaining traffic signal time t_{sig}^{remain} , both of which must meet minimum requirements. Here, we used the metric MAE as loss function and adam as an optimizer for our model which is shown in equation

Algorithm 2 Traffic Signal Aware EV Selection and Resource Allocation Algorithm

Input: $E_v^{max}, e_{v,r}^{arrive}, e_v^{need*}, t_{sig}^{remain}, n, E_{d,r}^{max}, e_{d,r}^{max}, e_{d,r}^{min}$
 $\forall v \in V, \forall r \in R$

Output: $\eta_{v,r}^{demand}, S_{v,r}$

Initialization: Neural Network Model Parameters
while (*Episode* $\leq$ *Maximum*) **do**

2: **for** $\forall v \in V$ **do**

if (Training == True) **then**

4: **Check constraint:** Calculate the constraints (16b), (16c), 16(j)
 Evaluate: Mean Absolute Error using (17)
6: **Store:** Trained Model Parameters

else

8: **Evaluate:** $\eta_{v,r}^{demand}$ from Trained Model
 Store: $\eta_{v,r}^{demand}$

10: **end if**

end for

12: **end while**

for $\forall v \in V$ **do**

14: **Evaluate constraint:** Calculate constraints (16d), (16e), (16f), (16g), (16h) and (16i)
 Initialization: Initially load = 0, count = 0
 while (*load* $\leq E_{d,r}^{max}$ and *count* $\leq n$) **do**

16: **if** ($\eta_{v,r}^{demand} \geq \text{MinVal}$) and ($t_{sig}^{remain} \geq \text{MinVal}$) **then**

Set: $S_{v,r} = 1, EV_{Selected}$

18: **Increment:** count = count + 1
 Update: Increase load as, load += $pow_{r,n}^{ch}$

20: **Store:** Corresponding EV ID, I_v

else

22: **Set:** $S_{v,r} = 0$, EV Not Selected

end if

24: **end while**

end for

17. The step by step procedure for EV selection and resource allocation process is shown in Algorithm 2.

V. EXPERIMENTAL RESULTS

For experiment, a road network with an area of $(6000 \times 4000)m^2$ is considered in which there are 15 nodes as intersection points and 22 edges connecting the nodes as road sections [32]–[34]. There are 3 types of two way roads having number of lanes of 6, 8 and 10 and the length of the individual edges are ranging from 800 to 1600 meter and the road network is designed according to the right-hand traffic rule [34]. We have considered one traffic intersection with 60 sec signal time and enable with wireless power transfer mechanism, and also equipped with dynamic EV selection and resource allocation process assisted by a base station. And, a total of 120 EVs are used for our experiment where there are 6 types of EVs considering the maximum battery capacity ranging from 5000 to 7500 watt and maximum speed ranging from 9 to 27 m/sec [34], [35]. Also, we have defined a total of 20 routes for the EVs in our system where the length of

the individual routes are ranging from 2300 to 7500 meter [34] and a total of 50 charging pads are used which are integrated with roads that can be switched on/off according to the output of the EV selection process of the deep learning models. It is worth mention-able that a number of EVs will be selected and allocated the resources while satisfying the maximum number of charging pads as well as under the maximum tolerable load capacity of the WPT system. We have used the simulation software SUMO (Version 1.17.0.1318) elaborated as Simulation of Urban Mobility, for road network design, EV modelling and generating the travel information and emission data generation for EVs in our work [34]–[36]. A road network environment scenario is simulated in SUMO so that EVs may halt in the WPT enabled traffic signal and, simulated and extracted the travel and emission data for all the EVs [37] and processed the data to feed into the deep learning models. For the implementation of deep learning models, we used PyCharm community edition as IDE with python 3.11 and several machine learning packages which includes tensorflow 2.13.0, keras 2.13.1, numpy 1.25.0, pandas 2.0.2, matplotlib 3.7.1, scikit-learn 1.2.2 etc. All the simulation and development process is conducted on a computer having 12th Gen Intel(R) Core(TM) i5-12500 CPU with 3.00 GHz processing frequency, 16GB of main memory and the machine is operated with 64-bit Windows 10 operating system. The simulation parameters used for the implementation design and performance evaluation process is shown in Table I.

TABLE I: SIMULATION PARAMETERS

Parameter Name	Value
Neural network	LSTM
Number of units	16, 100
Loss function	MAE
Optimizer	adam
Number of epocs	200
Batch size	8

We implemented our deep learning models based on LSTM for the prediction of battery energy need of the EVs to reach the destination place and for generating energy demand score in the current situation. The energy demand score mainly represents the importance of getting the charging resources based on the current battery status and the amount of battery energy need in present condition. In addition, conditional approach which ensures the traffic signal time and maximum capacity of the WPT system, is used to finally select the EVs and allocate resources. And, we also compared the numerical results of our models with the ground truth values as well as with the outputs of the multi-layer perceptron based deep neural network (DNN) model which we considered as a baseline model.

Fig. 3 illustrates the training and validation performance of DNN and LSTM model considering evaluation metric MAE for the battery energy need prediction task for 200 epochs [31], [38]. The model performance is evaluated based on the test dataset, and the performance results are shown in Table II whereas our proposed LSTM model achieves the lowest

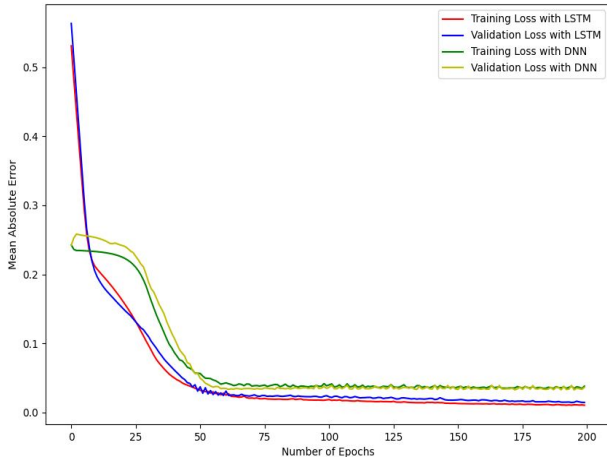


Fig. 3: Training and Validation loss for DNN and LSTM.

MAE error value of 0.014, whereas the DNN model produces a MAE value of 0.038. Fig. 4 depicts the predicted battery energy need values using the LSTM and DNN model where the LSTM model output represents better performance compared to baseline method. Also, Fig. 5 and Fig. 6 represents the performance of training and validation process and the generation of energy demand score using our proposed method and baseline method, respectively. The proposed LSTM model achieves best test MAE error value of 0.005 compared to the lowest test MAE value of 0.021 for the case of energy demand score generation while outputting more closer results to ground truth values compared to DNN model. The detail testing performance results for both the models is shown in Table II according to different epoch number and batch size.

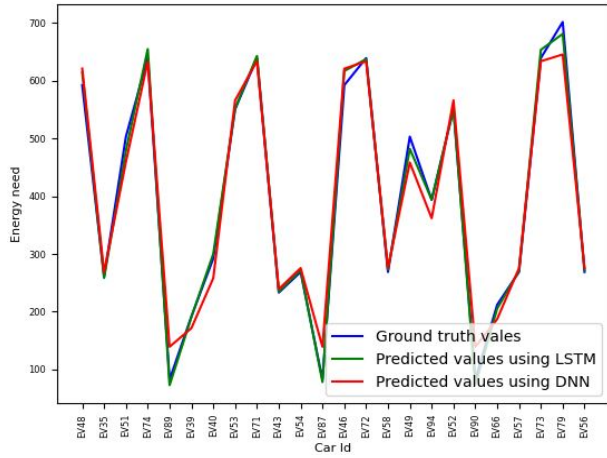


Fig. 4: Battery energy need values using DNN and LSTM.

Fig. 7 indicates the EV selection results for both models, where we measure the results by varying energy demand score and remaining signal time values and also presented the actual number of EVs as ground truth values under selection conditions. The LSTM model can correctly select the actual number of EVs for all the cases whereas the DNN model only gives correct output for the score value of 0.33 and failed to generate correct results for the score values of 0.34 and 0.35.

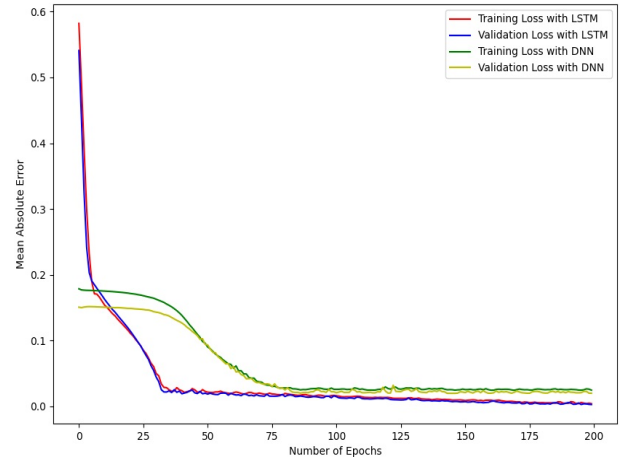


Fig. 5: Training and Validation loss for DNN and LSTM.

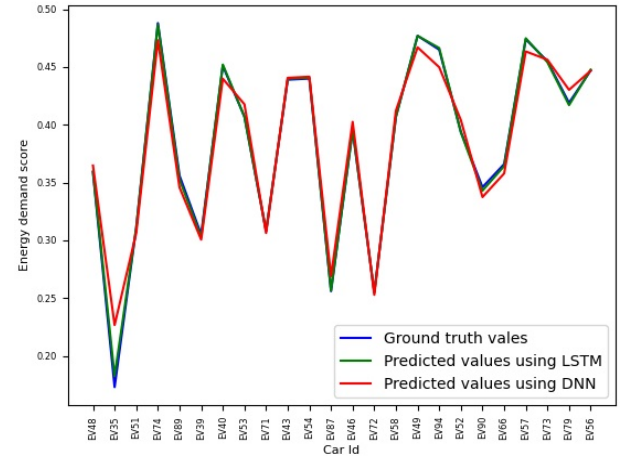


Fig. 6: Energy demand score values using DNN and LSTM.

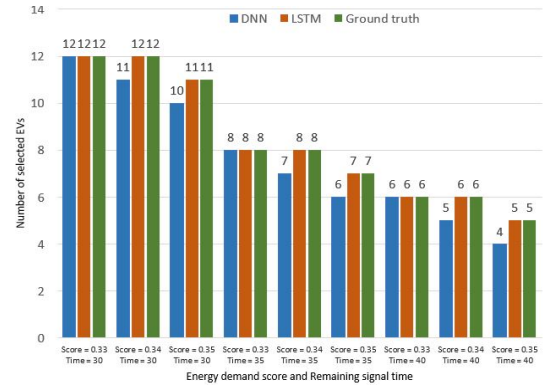


Fig. 7: Selection status of EVs using DNN, LSTM.

In addition, Fig. 8 shows the results of DNN and LSTM model for the EV selection and resource allocation process considering two cases where the minimum value of demand score and remaining signal time is considered as 0.33 and 35, and 0.35 and 30 respectively. In Fig.8, we presented two sub-figures, each containing three sections that depict EV

TABLE II: Performance results for DNN and LSTM model

NN Model	Batch Size	Battery energy need calculation (MAE values)			Energy demand score calculation (MAE values)		
		Epoch 100	Epoch 150	Epoch 200	Epoch 100	Epoch 150	Epoch 200
DNN	8	0.043	0.041	0.038	0.025	0.023	0.021
	16	0.046	0.044	0.04	0.026	0.022	0.021
	32	0.086	0.043	0.042	0.101	0.032	0.022
LSTM	8	0.024	0.018	0.014	0.014	0.007	0.005
	16	0.034	0.022	0.017	0.017	0.012	0.011
	32	0.047	0.035	0.021	0.019	0.018	0.014

information, such as EV ID, energy demand score, remaining signal time, output using the DNN model, and output using the LSTM model. In the first case, with a score value of 0.33 and a signal time value of 35, both models correctly selected the EVs. However, in the second case, the DNN model failed to select one EV that met the selection conditions, as indicated by the red rectangle.

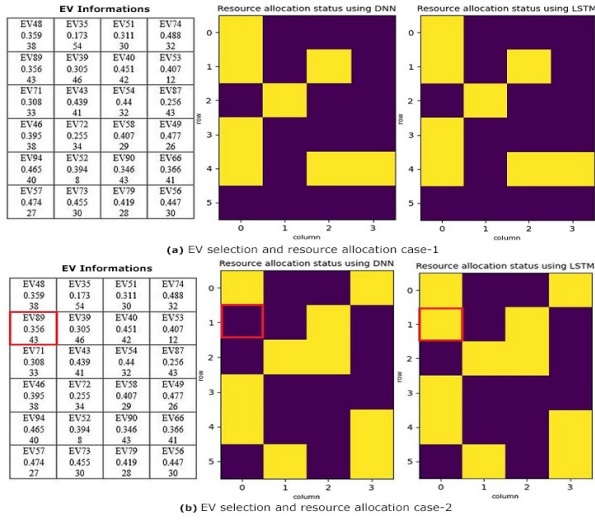


Fig. 8: EV selection and resource allocation.

We also measured the average obtained battery energy information for selected EVs by employing charging stations in SUMO and varying the remaining signal time, and we compared the results of the obtained energy from our proposed method with three baseline methods, baseline-A, baseline-B, and baseline-C, which are shown in Fig. 9. In case of baseline-A, it is considered that there is no option for WPT and hence the obtained energy values will be zero for all cases. And for baseline-B, we considered that the wireless power transfer is possible but in a static way and there is no option for selecting EVs according to the energy need. The term static for baseline-B indicates that all the EV will be charged from the grounded charging pads during traffic signal halt and so, the power is equally distributed to all the available pads according to maximum capacity of the node for facilitating charging. For our experiment, we have considered the selection of maximum 20 EVs among 50 charging points and the peak load for the WPT node is considered as 2000kW. As, EV selection is not possible for baseline-B, hence, we consider each charging pad

has the ability to transfer 40kW power per hour for static case. And this power transfer rate is fixed and so, the average obtained energy for baseline-B will remain static and will not increase with the time as shown in Fig. 9. And, for baseline-C and for our proposed method, dynamic EV selection and resource allocation is considered based on the DNN and LSTM model results. Remember that for both the cases, it is possible to dynamically select EVs and allocate them the charging resources and each charging pad has the ability to transfer 100kW of energy per hour [36]. So, for both the cases of baseline-C and our proposed method, we see from Fig. 9 that, the average obtained energy value for EVs increased with the increase of remaining signal time. Therefore, considering the above mentioned experimental results and analysis, the LSTM based model represents better performance than the DNN model for energy need prediction and energy demand score generation, and the LSTM based method ensures the effectiveness of our proposed system while achieving on average 55.1% and 8.4% more battery energy for EVs.

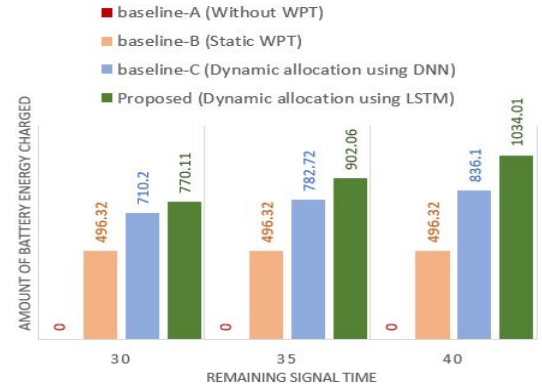


Fig. 9: Average obtained energy w.r.t. signal time.

VI. CONCLUSION

In this paper, we have introduced a novel framework empowered with artificial intelligence capability for selecting electric vehicles according to the current battery energy status and battery energy needed to reach destination in the present situation and allocating the selected EVs the charging resources. By using the proposed methodology it is possible to determine the energy need information of all the EVs and assign a energy demand score based on the need which in turn is considered as a criterion for selecting EVs along with remaining traffic signal time. For determining energy need values and generating demand scores, we utilized deep learning model based on LSTM. Depending on the demand score and signal time as conditions, EVs are selected, and resources are allocated. Finally, the experimental results ensure the ability of our proposed method for selecting EVs according to the requirement and allocating them the charging resources. The numerical results demonstrate that our proposed AI framework outperforms the DNN based model. Additionally, EVs can achieve 55.1% more battery energy compared to baseline-B method and 8.4% more energy compared to baseline-C method.

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